

SANKUR KUNDU

Software Engineer | Backend & Full-Stack

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SUMMARY

Software engineer (B.Tech CSE, 2026) who has designed, built and shipped a complete product alone. As co-founder of Gigzen: a **three-app delivery platform** on one PostgreSQL backend with **row-level security on every table**, in-database dispatch and Razorpay/UPI payments, plus **Populace**, an open-source testing tool on npm.

EXPERIENCE

Gigzen — Co-founder & Founding Engineer

Apr 2026 – Present

Bhubaneswar, India · Early-stage startup (incorporation in progress) · Sole engineer across every product

- Sole engineer on **Waggle**, a local-delivery platform of **three Android apps** (Waggle Gig for riders, Waggle Business for shops, Waggle for customers) built from one **TypeScript / React / Capacitor** codebase on a shared **Supabase PostgreSQL** backend: **53 tables, 90 RLS policies, 154 SQL functions**.
- Put the business rules in the database, where no client can bypass them: **first-accept-wins dispatch** with a **45-second offer window**, and **4-digit handover codes** gating pickup and delivery, so no job is double-assigned or skipped.
- Built payments and billing: **Razorpay** online payments, **UPI** rider settlement reconciled by UTR, numbered **GST invoices**, coupons, and month-end billing runs scheduled with **pg_cron**.
- Built real-time logistics: nearby-job matching, live rider tracking with **snap-to-road** map matching, voice turn-by-turn with rerouting, a distance-based live ETA, and parcel delivery (Waggle Send).
- **Security-tested** the platform as one system: **150 automated attacks** on the database access rules, all blocked; a **42-step Playwright** end-to-end run across all three apps, from rider verification to delivery rating.
- Earlier shipped **Waggle 1.4** (open source): **43 languages** across **92 countries**, offline-first messaging, phone numbers hidden at the database level.
- Built **Populace**, an open-source multi-user testing tool (**Node.js, zero runtime dependencies**) shipped as an **npm CLI**, a **GitHub Action** and an **Electron** desktop app. Its **first run found 5 defects in 3.5 minutes** after full manual testing had passed, including one that silently broke **every new signup**.
- Designed an app-agnostic **13-method adapter contract** with per-endpoint p50/p95 latency and three production-safety guards; proven on third-party software (**Gitea**: 1,106 API calls, zero failures); **122 self-tests** in CI.
- Correctness-tested under concurrency: **3.17M API calls** across seven runs of 30–300 simulated users with **zero API failures**, all 740 test accounts cleaned up and verified in the database; pinpointed throughput saturation (~200 users per machine).

Freelance Data Scientist — Upwork, remote (client in Spain)

Jun 2026 – Jul 2026

- Merged two Spanish hotel censuses (**5,909 and 8,058 records**; 118 and 281 columns) from Excel and STATA .dta sources, joined on national statistics identifiers with Python and Pandas.
- Built a Google Trends panel of **10,038 city-months**: deseasonalised, z-scored and flagged for positive demand shocks.

PROJECTS

quorumdb — Rust, zero dependencies: a distributed SQL database, PostgreSQL wire-compatible

- Built from scratch a **distributed SQL database** that **psql** connects to: an **LSM storage engine**, **Raft consensus**, a **sharded multi-Raft store** that splits and rebalances itself, **serializable ACID transactions** (Percolator) and a SQL planner; **1,500 tpmC** on a **TPC-C**-derived benchmark, consistency checks passing.
- Proved correct by **deterministic simulation** (**10,000 Raft clusters, 610K crashes, linearizability** and **serializability** machine-checked) and a **TLA+** model checked by TLC over **337K states**; found **6 real bugs**, catches all **15 planted ones**.
- Built a **time-travel debugger** that steps a simulated Raft cluster event by event and rewinds it by replay.

ML Model Serving Platform — Python, Flask, Redis, NVIDIA Triton, Docker, Kubernetes

- Inference service with versioned rollout, A/B traffic splitting, circuit breaking and drift detection. Fixed **six defects**, each pinned by a regression test (**65 tests** in CI), including an **authentication bypass** and a **non-atomic Redis rate limiter**.

SKILLS

CS Fundamentals: Data Structures & Algorithms, System Design, Distributed Systems (Raft, MVCC, 2PC), DBMS internals (LSM trees), OS, Computer Networks

Languages: Rust, TypeScript, JavaScript, Python, SQL (PL/pgSQL), Java

Backend & Data: PostgreSQL, Supabase, Row-Level Security, pg_cron, REST APIs, Node.js, FastAPI, Flask, Redis, Razorpay/UPI

Frontend & Mobile: React, Capacitor (Android), Electron, Leaflet/MapLibre maps, HTML, CSS

Testing, DevOps & Cloud: Deterministic simulation testing, TLA+ model checking, Playwright, pytest, GitHub Actions CI/CD, Docker, Kubernetes, GCP, AWS, Git, Linux

EDUCATION

B.Tech, Computer Science and Engineering (AI & ML)

Institute of Engineering and Management (IEM), Kolkata

Sep 2022 – Apr 2026